

ACF for the AR(2) Process : Yule-Walker

General solution (distinct roots)

$$\rho(h) = C_1 \lambda_1^{|h|} + C_2 \lambda_2^{|h|}$$

$$\rho(0) = 1 \Rightarrow C_1 + C_2 = 1 \quad (1)$$

$$\rho(1) = \frac{\phi_1}{1 - \phi_2} \Rightarrow C_1 \lambda_1 + C_2 \lambda_2 = \frac{\phi_1}{1 - \phi_2} \quad (2)$$

Auxiliary equation:

$$y^2 - \phi_1 y - \phi_2 = 0$$

$$\Rightarrow (y - \lambda_1)(y - \lambda_2) = 0$$

$$\Rightarrow y^2 - (\lambda_1 + \lambda_2)y + \lambda_1 \lambda_2 = 0$$

$\therefore$ Equating coefficients of y :

$$\phi_1 = \lambda_1 + \lambda_2$$

$$\phi_2 = -\lambda_1 \lambda_2$$

$$\therefore \text{In } (2) \quad C_1 \lambda_1 + C_2 \lambda_2 = \frac{\lambda_1 + \lambda_2}{1 + \lambda_1 \lambda_2}$$

$$\text{From } (1): \quad C_2 = 1 - C_1$$

$$\Rightarrow C_1 \lambda_1 + (1 - C_1) \lambda_2 = \frac{\lambda_1 + \lambda_2}{1 + \lambda_1 \lambda_2}$$

$$c_1(\lambda_1, -\lambda_2) = \frac{\lambda_1 + \lambda_2}{1 + \lambda_1 \lambda_2} - \lambda_2$$

$$= \frac{\lambda_1 + \lambda_2 - \lambda_2(1 + \lambda_1 \lambda_2)}{1 + \lambda_1 \lambda_2}$$

$$= \frac{\lambda_1(1 - \lambda_2^2)}{1 + \lambda_1 \lambda_2}$$

$$\Rightarrow c_1 = \frac{\lambda_1(1 - \lambda_2^2)}{(\lambda_1 - \lambda_2)(1 + \lambda_1 \lambda_2)}$$

$$c_2 = 1 - c_1$$

$$= \frac{-\lambda_2(1 - \lambda_1^2)}{(\lambda_1 - \lambda_2)(1 + \lambda_1 \lambda_2)}$$

$$e(h) = c_1 \lambda_1^{|h|} + c_2 \lambda_2^{|h|}$$

$$= \frac{\lambda_1(1 - \lambda_2^2) \lambda_1^{|h|} - \lambda_2(1 - \lambda_1^2) \lambda_2^{|h|}}{(\lambda_1 - \lambda_2)(1 - \lambda_1 \lambda_2)}$$

Repeated Roots

$$e(h) = [c_1 + c_2 |h|] \lambda^{|h|} \quad \lambda_1 = \lambda_2 = \lambda$$

$$e(0) = 1 \Rightarrow c_1 = 1$$

$$h = 1: \quad (c_1 + c_2) \lambda = e(1) \\ = \frac{\phi_1}{1 - \phi_2}$$

$$(1 + c_2) \lambda = \frac{\phi_1}{1 - \phi_2}$$

$$\begin{aligned} \phi_1 &= \lambda_1 + \lambda_2 & \phi_2 &= -\lambda_1 \lambda_2 \\ &= 2\lambda & &= -\lambda^2 \end{aligned}$$

$$\therefore (1 + c_2) \lambda = \frac{2\lambda}{1 + \lambda^2}$$

$$1 + c_2 = \frac{2}{1 + \lambda^2}$$

$$\Rightarrow c_2 = \frac{1 - \lambda^2}{1 + \lambda^2}$$

$$\therefore e(h) = \left[1 + \frac{1 - \lambda^2}{1 + \lambda^2} |h| \right] \lambda^{|h|}$$

Complex Roots

λ_1, λ_2 will be complex conjugates of each other.

$$\lambda_1 = \lambda e^{i\theta}; \quad \lambda_2 = \lambda e^{-i\theta}.$$

$$\begin{aligned}\phi_1 &= \lambda_1 + \lambda_2 \\ &= \lambda (e^{i\theta} + e^{-i\theta}) \\ &= 2\lambda \cos \theta\end{aligned}$$

$$\begin{aligned}\phi_2 &= -\lambda_1 \lambda_2 \\ &= -\lambda^2\end{aligned}$$

$$e(h) = c_1 \lambda_1^{|h|} + c_2 \lambda_2^{|h|}$$

c_1 and c_2 must be complex conjugates of each other.

$$\text{Let } c_1 = R e^{i\phi}; \quad c_2 = R e^{-i\phi}$$

$$\begin{aligned}\therefore e(h) &= R e^{i\phi} [\lambda e^{i\theta}]^{|h|} + R e^{-i\phi} [\lambda e^{-i\theta}]^{|h|} \\ &= R \lambda^{|h|} e^{i(\phi + \theta|h|)} + R \lambda^{|h|} e^{-i(\phi + \theta|h|)} \\ &= R \lambda^{|h|} [e^{i(\phi + \theta|h|)} + e^{-i(\phi + \theta|h|)}] \\ &= 2R \lambda^{|h|} \cos(\phi + \theta|h|).\end{aligned}$$